Paleomagnetism of the Great Slave Supergroup, Northwest Territories, Canada: multicomponent magnetization of the Kahochella Group

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Results are reported from 59 stratigraphic horizons spanning an aggregate of some 2000 m of sedimentary rocks of the Great Slave Supergroup, mostly located in the Kahochella Group. Partial demagnetization experiments and vector differences indicate that the magnetization involved is multicomponent, with a well-defined overprint masking an underlying magnetic component. The overprint has now been recognized in many formations of the Great Slave Supergroup and is interpreted as having been acquired in response to uplift and cooling following orogeny in the Coronation Geosyncline. It corresponds to a paleomagnetic pole at 91°W, 21°N ($A_{95} = 5°$). Removal of this overprint reveals the underlying, presumed primary, magnetization, which has a mean direction of $D = 129°$, $I = +21°$ ($N = 18$ sites, $k = 12$, $\alpha_{95} = 10°$) and a corresponding pole at 62°W, 7°S ($K = 15$, $A_{95} = 9°$). The ages of these poles cannot be fixed precisely, but the available data suggest that the "primary" remanence was acquired about −1.8 Ga, and the overprint about −1.7 Ga. The two poles thus help define the Precambrian polar wander curve for Laurentia. In particular they help define the so-called Coronation loop, which reflects the response of Laurentia to the major orogeny in the Coronation Geosyncline.

On rapporte les résultats pour 59 horizons stratigraphiques qui recoupent au total quelques 2000 m de roches sédimentaires dans le supergroupe de Great Slave, localisées surtout dans le groupe de Kahochella. Les expériences de désaimantation partielle et les différences entre les vecteurs indiquent que l'aimantation de ces roches possède plusieurs composantes avec une surimpression bien définie qui masque une composante magnétique sous-jacente. On a maintenant reconnu cette surimpression dans plusieurs formations du supergroupe de Great Slave et on l'interprète comme ayant été acquise lors du soulèvement et du refroidissement qui ont suivi l'orogénèse dans le géosynclinal de Coronation. Elle correspond à un pôle paléomagnétique à 91°O, 21°N ($A_{95} = 5°$). La séparation de cette surimpression révèle l'aimantation sous-jacente, présumément primaire, qui a une direction moyenne de $D = 129°$, $I = +21°$ ($N = 18$ sites, $k = 12$, $\alpha_{95} = 10°$) et un pôle correspondant à 62°O, 7°S ($K = 15$, $A_{95} = 9°$). On ne peut déterminer avec précision l'âge de ces pôles mais les données disponibles suggèrent que la rémanence "primaire" a été acquise il y a environ 1,8 Ga et la surimpression il y a 1,7 Ga environ. Les deux pôles aident ainsi à définir la courbe de migration polaire de la Laurentie au Précambrien. En particulier, ils aident à définir la soi-disant boucle de Coronation qui reflète la réponse de la Laurentie à l'orogénèse majeure dans le géosynclinal de Coronation.

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Geology and sampling

The rocks of the Athapuscow Aulacogen form a major remnant of the Coronation Geosyncline (Hoffman *et al.* 1970; Hoffman 1973), which occurs on the edges of the Slave Province. Rocks of the aulacogen are exposed in the East Arm of Great Slave Lake and form in part the Great Slave Supergroup consisting of more than 8 km of sandstones, lavas, red and green shales, and stromatolitic limestone and dolomite. The Kahochella Group consists mainly of unmetamorphosed mudstones and siltstones and comprises one of four groups that together make up the Great Slave Supergroup, which is unconformably overlain by the Et-Then Group (red and buff conglomerates, shales, siltstones, and sandstones). In the area sampled the Kahochella Group conformably overlies the Seton Formation (red shales and spilitized basalts). The Seton Formation has been dated by Baadsgaard *et al.* (1973) using the Rb–Sr method, and by Cumming (1980) using lead isotopes. Cumming also quotes unpublished results by the Geological Survey of Canada, and recalculates the ages using the decay constants recommended by Steiger and Jager (1977) ($\lambda^{87}Rb = 1.42 \times 10^{-11}$ year^{-1}, $\lambda^{238}U = 1.55125 \times 10^{-10}$ year^{-1}, $\lambda^{235}U = 9.8485 \times 10^{-10}$ year^{-1}). He obtains Rb–Sr ages of 1833 ± 28 and 1805 ± 16 Ma, whilst his own Pb/Pb work yields an age of 1804 ± 23 Ma. The agreement between the methods is excellent and thus the date obtained probably corresponds to the initial formation of the Seton lavas. McGlynn *et al.* (1974) suggest that the Great Slave Supergroup was buried, folded, and faulted, subsequently being uplifted and cooling at about −1.7 to −1.6 Ga (see also McGlynn and Irving 1978; Irving and McGlynn 1979).

The Kahochella Group was sampled on Keith Island (111.9°W, 62.1°N, Fig. 1) where the strata dip at about 15° to the north and northeast. Samples were collected from 49 sites in the Kahochella Group and 5 sites each in the Seton Formation below and the Pethei Group above. The sites were grouped in six sections, lettered A–F, within each of which the between-site spacing was about

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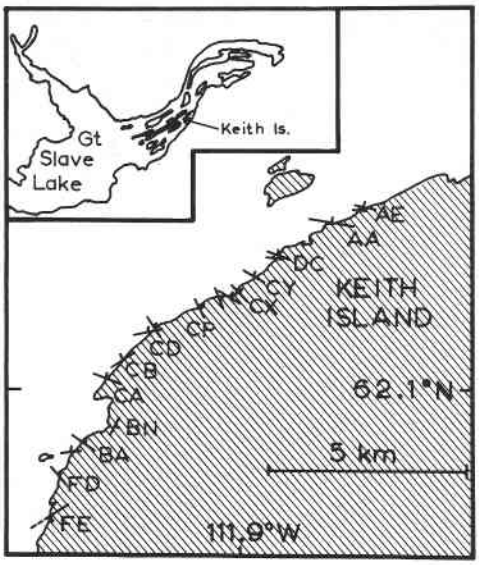


FIG. 1. Location of sampling sites on northwest shore of Keith Island in Great Slave Lake, N.W.T., Canada. The strike of the strata is marked at each site. Unmarked sites lie in alphabetic sequence between the marked ones.

25 m stratigraphically. The larger gaps between sections were forced by lack of exposure or unsuitable lithology.

Section A is in the biogenic, primarily calcareous rocks of the Pethei Group. These samples gave poor results and are included here only for the sake of completeness. Sections B and C are in the Gibraltar, McLeod Bay, and Charlton Bay Formations. Site CZ was sampled in a small dioritic laccolith intruding the Charlton Bay Formation. Section D is the baked contact of the intrusion, thought to be about 10 m thick. Site DA is within a few cm of the contact and sites DB, DC, and DD are 3–6, 50, and 100 m from the contact. Section E is a detailed section between two sites (CE and CF) in the C section at a stratigraphic spacing of about 4 m; it was sampled in order to examine possible geomagnetic field changes missed in the longer intervals between sites in the C section. Section F is sampled in the intercalated red shales and lavas of the Seton Formation.

Four samples, independently oriented by means of a sun compass, were obtained at each site, usually by field drilling. Where the fissile nature of the rock prevented field drilling, hand samples were taken. As far as possible all samples within a site were obtained from the same stratigraphic horizon. Two specimens 25 mm in length were cut from each 25 mm diameter core sample.

Results

The natural remanent magnetization (NRM) of the specimens (typically $5–10 \times 10^{-6}$ $A \cdot m^2 \cdot kg^{-1}$, i.e., $1.5–3.0 \times 10^{-5}$ emu/cm^3) was found to have a significant unstable component, leading to changes in direction of up to 30° over 6 months' storage. Partial demagnetization was therefore imperative.

Alternating field (AF) demagnetization

A pilot specimen from each site was subjected to alternating field demagnetization up to the limit of the apparatus (peak flux density 180 mT = 1800 Oe). Curves of normalized intensity against demagnetizing field exhibit sharp drops at fields less than 20 mT (Fig. 2) revealing the existence of a large soft magnetization in specimens from all sections except A. The soft magnetization was usually removed by treatment in peak fields of 40 mT. In addition to the pilot specimens, demagnetization of one specimen per sample at fields of 40, 60, and 80 mT was undertaken to provide evidence of demagnetization end points. This was followed by treatment in higher fields in cases where no end point had been reached at or before 80 mT. The results giving maximum precision at each site were taken to represent the best cleaned direction unless all the specimen directions moved coherently after attainment of maximum precision. In these cases an end point was chosen where successive demagnetizations produced no further coherent change. Site level statistics after AF demagnetization are shown in Table 1. The directions form a well-defined group, away from the earth's field, indicating the presence of a stable magnetization.

Thermal demagnetization

Since one of the major magnetic minerals in the rocks is hematite, which can have very high coercivities (Roquet 1954), thermal demagnetization was also undertaken. Limited initial studies were carried out at the Earth Physics Branch, Department of Energy, Mines and Resources laboratories in Ottawa. The bulk of the work employed the University of Alberta thermal demagnetization equipment, which is located inside a magnetically shielded room with residual field over the

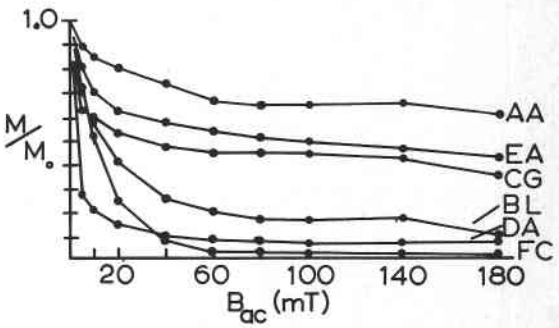


FIG. 2. Alternating field demagnetization curves. The vertical axis is intensity of magnetization normalized to the NRM value, on a linear scale. The horizontal axis is peak alternating flux density in milliteslas.

TABLE 1. Site mean directions after AF and thermal cleaning. Dip correction applied

	AF demagnetization					Thermal demagnetization							
Site	*N*	mT	*D*	*I*	*k*	*N*	°C	*D*	*I*	*k*	Strike	Dip	Group
AA	4	100	175	61	24	3	550	199	47	47	282	14	
AB	4	140	140	63	284	4	500	139	63	29	280	13	
AC	4	20	20	73	320	Random					280	13	
AD	4	10	149	49	191	4	500	154	45	165	277	12	
AE	4	80	119	62	72	4	500	109	68	6	286	17	
BA	3	140	276	15	34	3	630	282	2	60	307	18	2r
BB	4	100	276	79	10	4	500	353	80	6	288	15	?
BC	4	40	352	80	102	4	500	73	82	31	251	17	1
BD	4	100	99	75	326	4	500	92	74	91	297	20	1
BE	4	140	276	67	6	Random					285	22	?
BF	4	40	299	87	9	3	550	1	40	16	245	14	?
BG	4	60	152	58	919	4	550	141	62	1097	318	12	1
BH	4	60	253	12	11	4	630	283	−22	26	3	18	2r
BI	3	40	138	59	114	3	500	129	64	98	0	16	1
BJ	Random					Random					346	15	?
BK	4	100	331	1	13	4	600	328	−8	37	2	12	2r
BL	3	5	141	72	265	3	300	147	60	117	4	14	1
BM	4	10	217	89	61	4	600	340	16	19	14	13	2r
BN	4	5	142	87	32	4	660	328	−31	35	27	11	2r
CA	3	40	160	39	58	3	550	164	34	223	294	7	2n
CB	4	80	146	56	23	4	600	146	45	12	237	21	2n
CC	4	40	164	72	42	4	500	126	55	32	263	7	1
CD	4	40	119	56	135	4	550	112	56	32	330	13	1
CE	4	5	129	53	109	3	500	142	35	222	6	14	1
CF	4	80	149	49	634	5	600	148	42	181	11	14	1
CG	4	60	139	44	721	4	550	136	39	122	23	20	1
CH	4	180	138	47	176	4	547	134	36	153	16	17	1
CI	4	40	143	59	87	3	630	311	32	33	354	15	2r
CJ	4	100	80	26	176	4	660	82	17	305	338	32	2n
CK	3	80	126	39	202	3	660	124	30	144	318	13	2n
CL	4	80	125	34	319	4	625	123	19	613	332	18	2n
CM	4	60	135	42	103	4	600	134	25	93	334	14	2n
CN	4	40	150	70	65	4	660	300	4	55	330	15	2r
CO	4	80	276	28	48	4	600	292	−7	89	337	16	2r
CP	4	80	146	54	272	4	630	145	42	53	334	13	1
CQ	4	80	124	49	13	Random					347	20	2m
CR	4	60	128	54	345	4	550	116	45	39	338	14	1
CS	4	80	136	58	31	4	500	134	52	20	336	15	1
CT	4	80	137	51	62	4	550	137	29	92	332	16	2n
CU	4	60	317	−17	21	4	600	315	−29	60	302	10	2r
CV	4	40	128	56	438	3	660	129	40	172	304	10	2n
CW	4	180	119	40	85	5	500	123	40	117	305	14	2n
CX	4	60	148	74	78	4	550	147	71	186	315	12	1
CY	4	40	155	69	132	4	500	132	74	357	303	12	1
CZ	4	5	9	70	48	No result					278	20	?
DA	5	40	82	81	49	4	400	111	60	110	278	20	1
DB	3	10	21	79	167	5	300	105	82	103	278	20	1
DC	4	60	123	76	106	5	500	113	70	243	278	20	1
DD	4	60	163	69	221	4	500	156	65	174	292	13	1
EA	4	80	121	50	336	4	550	122	47	273	353	18	1
EB	4	80	136	53	743	4	550	124	47	546	353	18	1
EC	4	100	129	51	295	4	550	126	47	789	353	18	1
ED	4	80	143	60	38	4	550	144	60	48	1	22	1

TABLE 1. (*Concluded*)

Site	AF demagnetization					Thermal demagnetization					Strike	Dip	Group
	N	mT	*D*	*I*	*k*	*N*	°C	*D*	*I*	*k*			
EE	4	40	137	52	473	4	600	146	42	103	1	22	1
FA	4	60	24	50	53	4	500	20	46	24	270	37	1
FB	4	5	348	50	454	2	550	351	64	101	270	37	1
FC	4	10	70	64	70	4	550	85	62	74	319	22	1
FD	4	40	122	67	190	4	550	123	67	142	319	22	1
FE	4	20	169	44	582	4	550	150	23	162	58	38	1

N = number of specimens.
D = declination in degrees clockwise from north.
I = inclination in degrees positive downwards.
k = Fisher's precision parameter.
mT = peak demagnetizing field in milliteslas.
°C = peak demagnetizing temperature in degrees Celsius.
Strike = strike of strata, 90° counterclockwise from dip direction.
Dip = dip of strata.
Group = type of magnetization. 1 denotes complete remagnetization. 2 denotes an earlier remanence with either normal (n), reversed (r), or mixed (m) polarity within site.

heated space of less than 5 nT and a cycle time of approximately 1 h.

Pilot specimens from each site were subjected to stepwise thermal demagnetization. These studies revealed small but systematic differences between AF and thermally cleaned directions and blocking temperatures distributed from room temperature up the Néel point of hematite. It was concluded that there were at least two magnetizations present and that they were not well separated by AF cleaning. Thermal demagnetization of one specimen per sample was therefore undertaken at temperatures of 500, 550, 600, 630, and 660°C to obtain two steps below the Curie point of pure magnetite and some detail between this and the Néel point of hematite. Representative pilot thermal demagnetization curves are shown in Fig. 3, which clearly illustrates the distributed nature of the blocking temperature spectra. Some specimens exhibit significant intensity increases at high temperatures probably due to chemical changes caused by the heating. The basalts of section F show typical magnetite Curie points. The rationale behind the two groups indicated in Fig. 3 is explained below.

The site level statistics after thermal demagnetization and dip correction are shown in Table 1. The best cleaned directions were chosen on the same basis as that used for the AF results. The site mean directions are illustrated in Fig. 4. It will be noted that nine sites are approximately reversed with respect to the main group and show much greater scatter.

The rocks of the A section, sampled in a biogenic, primarily calcareous formation of the Pethei Group much higher in the succession, show coherent polarity but poor grouping. Thermal demagnetization appeared to remove a present field component from these rocks but the within-site grouping decreased in all cases, so that site AC failed Watson's (1956) test for randomness. Results from the A section are not discussed further. The results from the B and C sections are complex, suggesting more than one magnetization episode. The baked sites, DA and DB, have mean directions little different from their unbaked neighbours. This could be taken as evidence for a short period of time between the magnetization of the Kahochella Group and the emplacement of the intrusion, or for a general remagnetization episode. Results from the detailed E section are little different from those of the neighbouring sites, CE and CF, and reveal no short time scale features. Table 2 (entries 9 and 10) shows that the precision of the F section mean (which includes results from spilitized lavas) is significantly reduced by dip correction (McElhinny 1964), demonstrating that these sites were remagnetized after folding.

Discussion

Secondary magnetization

Figure 5 shows the site mean directions for the C section after AF and thermal treatments with no dip correction applied. The arrows point in the direction of the means after thermal treatment. A magnetization with a steep south-southeast direction has been consistently removed, suggesting that thermal demagnetization has been more effective in removing a secondary component. The directions for CO and CU have moved to reversed positions while two more reversed sites (CI and CN), thought to be normal after AF demagnetization, have emerged. The existence of such reversed sites demonstrates the retention of an earlier magnetization.

The secondary magnetization was further investi-

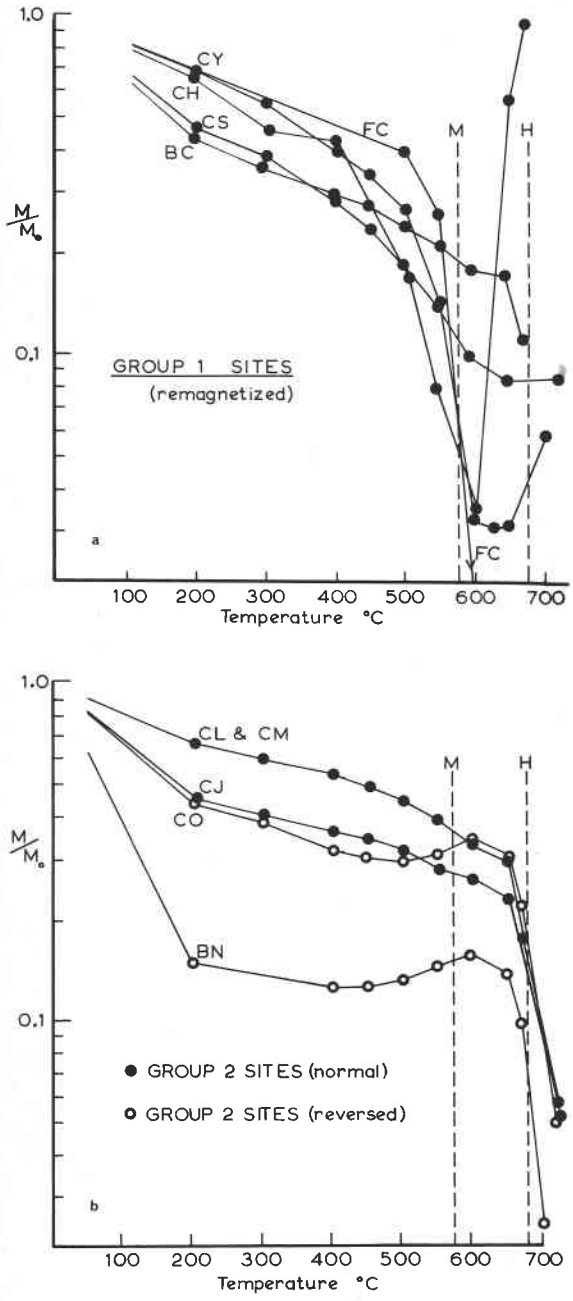


FIG. 3. Thermal demagnetization curves for samples from remagnetized (group 1) and partially remagnetized (group 2) sites. The vertical axis is intensity of magnetization normalized to NRM intensity on a logarithmic scale. The horizontal axis is peak oven temperature. The Curie points of magnetite (M) and hematite (H) are marked.

gated by calculation of the vector removed by each step of thermal demagnetization. A variety of directions was observed, but a coherent magnetization in a south-southeast direction with steep inclination was removed over the range 550–600°C from specimens with both

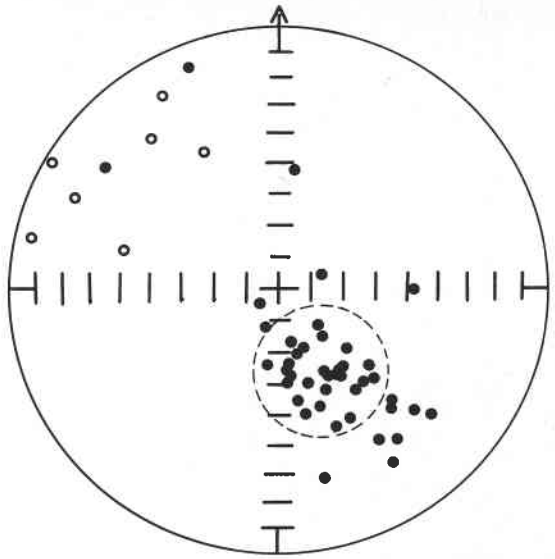


FIG. 4. Site mean directions after thermal demagnetization. The dotted circle marks the area within which 95% of the remagnetized population (as derived from removed vector studies) is expected to lie. Open circles: upper hemisphere. Dots: lower hemisphere. Schmidt equal area projection. The primitive plane is the present horizontal.

normal and reversed polarities. The results are given in Table 3, grouped by site. All nonrandom sites are included but sites from the closely spaced E section were omitted to avoid bias. For the sake of completeness, however, the E section results are included in Table 3, inspection of which shows that the sites involved also yield difference vectors close to the overall group. Between-site statistics for the removed vectors are given in Table 4, entry 1. Two of the 39 sites (CB and CD) have site mean removed vectors well away from the group and have less than a 1% chance of belonging to the same population as the rest of the observations. They have therefore been omitted and the mean direction has been recalculated (Table 4, entry 2). The result is a tightly grouped population of site mean removed vectors ($k = 45$). This is before a correction has been applied for geologic dip. Such a correction reduces k to 17. This change is significant at 99% confidence (McElhinny 1964), showing that the partial remagnetization episode occurred after folding. We first reported this overprint some years ago in a preliminary discussion (McMurry *et al.* 1973) and it has subsequently been found in several formations of the Great Slave Supergroup (see Evans and Bingham 1976; Bingham and Evans 1976; McGlynn and Irving 1978; Irving and McGlynn 1979; Evans *et al.* 1980).

The removed vector site mean population obtained in the present study has been tested for conformity to Fisher's (1953) distribution. The azimuthal distribution gives a χ^2 value of 2.73 with three degrees of freedom

TABLE 2. Between-site statistics on thermally demagnetized site means

Entry	N	D	I	k	α_{95}	Population
1	9	138	34	21	11	Group 2n (Table 1)
2	9	128	33	15	14	Group 2n, dip corrected
3	9	310	−14	8	19	Group 2r
4	9	308	−5	8	20	Group 2r, dip corrected
5	18	134	24	10	11	2n–2r
6	18	128	19	8	13	2n–2r, dip corrected
7	11	156	62	66	6	Site mean within 10° of removed vector
8	11	132	63	28	9	Entry 7 above, dip corrected
9	5	151	74	30	14	F section (Seton Fm, lavas and shales)
10	5	86	69	4	42	F section, dip corrected
11	20	152	59	48	5	Group 1 (Table 1)
12	20	132	59	23	7	Group 1, dip corrected
Secondary magnetization vectorially subtracted from sites of entries 1–6 above (see Table 5 for individual site means)						
13	9	135	26	14	14	Group 2n
14	9	128	24	11	16	Group 2n, dip corrected
15	9	314	−27	13	15	Group 2r
16	9	310	−19	12	16	Group 2r, dip corrected
17	18	134	26	14	9	2n–2r
18	18	129	21	12	10	2n–2r, dip corrected

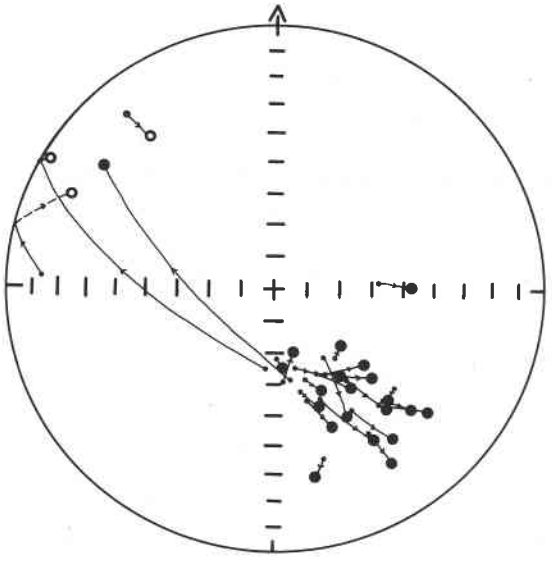


FIG. 5. Partial removal of secondary magnetization from rocks of the C section. Small symbols: site mean directions after AF treatment. Large symbols: site mean directions after thermal treatment. Circles: upper hemisphere. Dots: lower hemisphere. Schmidt equal area projection. The primitive plane is the present horizontal.

(six cells were used). The critical value is 7.8, so that we may not reject the hypothesis that the azimuthal distribution is uniform with 95% confidence. The angular deviation from the mean was tested using the maximum likelihood estimate of κ, (i.e., $k = N/(N - R)$) given by Watson (1956). This estimate of κ is the appropriate value for goodness of fit testing (McFadden 1980).

A value of 44.7 for κ and six cells of equal probability give a χ^2 value of 5.0, which is less than the critical value of 5.99 for two degrees of freedom. Thus at the 95% confidence level the null hypothesis that the angular deviation from the mean is Fisher distributed may not be rejected.

Examination of Table 1 shows that many of the sites reached maximum precision at demagnetization temperatures below 600°C. Thus the magnetization at many of these sites is likely to be contaminated by the remagnetization episode, which was identified on the basis of 550–600°C vector differences. All the thermally cleaned site mean directions were therefore tested to see whether they had a greater than 5% chance of belonging to the remagnetized population derived from the removed vector site means (see Fig. 4). The test is simply to compare the angle between each site mean and the remagnetization direction with θ_{95} (the semiangle of the one expected to contain 95% of the population) for the removed vector population.

The resultant division of sites into those not distinguishable (group 1) and those distinguishable (group 2) from the remagnetization is shown in the last column of Table 1. Group 1 is considered to be completely remagnetized, whereas group 2 is thought to retain, at least in part, an earlier magnetization.

A fold test applied to the group 1 site means

TABLE 3. Within-site statistics on the vectors removed by the thermal demagnetization step from 550 to 600°C, before dip correction

Site	N	D	I	k	α_{95}
BA	4	215	59	22	20
BB	4	158	63	35	16
BC	4	153	63	88	10
BD	4	151	63	236	6
BG	4	145	59	130	8
BH	4	186	57	37	15
BI	5	145	66	96	8
BK	2	165	67	226	17
BM	4	176	82	129	8
BN	4	153	64	67	11
CA	4	175	59	4	49
CB	4	58	57	5	43
CC	4	136	46	34	16
CD	3	98	42	12	37
CF	5	165	57	136	7
CG	4	157	54	58	12
CI	4	164	53	51	13
CJ	4	126	59	55	12
CK	3	141	48	13	36
CL	4	155	48	97	9
CM	4	156	58	20	21
CN	4	155	52	33	16
CO	4	172	51	101	9
CP	4	149	55	322	5
CQ	4	146	58	65	11
CR	4	146	49	24	19
CS	4	153	62	99	9
CT	4	103	65	15	24
CV	4	170	65	5	48
CX	4	166	66	50	13
CY	4	154	64	130	8
DB	2	152	64	141	21
DC	4	134	53	146	8
DD	3	137	58	137	11
FA	4	141	76	47	14
FB	2	203	72	50	36
FC	4	144	70	8	34
FD	4	160	66	32	16
FE	4	156	58	93	10
EA	4	144	60	240	6
EB	4	151	60	216	6
EC	4	152	57	342	5
ED	4	182	63	60	12
EE	4	164	60	436	4

demonstrates that their precision is significantly decreased at 95% confidence by applying a dip correction (Table 2, entries 11, 12). Thus the group 1 magnetization is postfolding in age. In addition, 11 of these sites in the B and C sections had mean directions and mean removed vectors within 10° of each other, which is strongly suggestive of only one magnetization at these sites. The precision of these sites on their own is significantly decreased (Table 4, entries 4, 5) by applying a dip correction. Their magnetization is thus almost entirely secondary and illustrates the very tight grouping ($k = 96$) of the remagnetization.

There remains a group of sites for which the hypothesis of complete remagnetization is rejected. The group contains 18 sites forming two groups of 9 each, which are approximately reversed with respect to each other (groups 2n and 2r in Tables 1, 2). The reversal is not exact, although an F test (Watson and Irving 1957) indicates that the hypothesis that they are antiparallel cannot be rejected with 95% confidence. However, from Fig. 4 it is evident that the two groups and the remagnetization direction are coplanar, with the reversed group more scattered than the normal group. These observations can be simply explained by the hypothesis that both groups still retain a component of the remagnetization vector. Figure 6 shows the three vectors involved. It may be seen that the secondary magnetization has a component directed against the reversed direction and along the normal direction. The remagnetization is well defined compared with that of either the 2n or 2r group. This well-defined component has the effect of reducing the length of the reversed vector without changing the orthogonal scatter component, thus increasing the angular scatter in the reversed vectors. Similarly the scatter in the normal direction is reduced.

Given the three observed directions, the original direction may be recovered geometrically if it is assumed that the normal and reversed groups should be exactly antiparallel and that the rock magnetic properties of normally and reversely magnetized sites are similar, so that their original intensities of magnetization were similar. This latter assumption is supported by the fact that the normal and reversed sites are stratigraphically intermingled (at least seven reversals are implied). From the angles it may be deduced that on average a remagnetization vector of length 30% of the primary vector has been added. This vector is easily removed mathematically. The operation must be carried out on the site mean directions before a dip correction is applied, because the remagnetization episode has been shown to be postfolding. The new corrected site directions before and after dip correction are given in Table 5. The means for the normal and reversed groups appear in Table 2 (entries 13, 15). Now that they are antiparallel their dispersions are the same. This supports the proposed model, because we have no reason to expect different dispersions for normal and reversed magnetizations.

The standard method of dealing with normal and reversed groups that are not quite antiparallel is to

TABLE 4. Between-site statistics on the vectors removed by the thermal demagnetization step from 550 to 600°C

Entry	N	D	I	k	α_{95}	θ_{95}	Population
1	39	151	62	29	4	26	Sections B, C, D, F. All nonrandom sites
2	37	155	62	45	4	21	Entry 1, but excluding CB, CD (see text)
3	37	130	61	17	6		Entry 2, dip corrected
4	11	150	60	96	5		Site mean within 10° of removed vector
5	11	128	60	39	7		Entry 4, dip corrected

N = number of site mean removed vectors.
D = mean declination.
I = mean inclination.
k = estimate of Fisher's precision parameter = $(N - 1)/(N - R)$.
α_{95} = angular radius of cone of 95% confidence about mean direction.
θ_{95} = angular radius of cone containing 95% of the population.

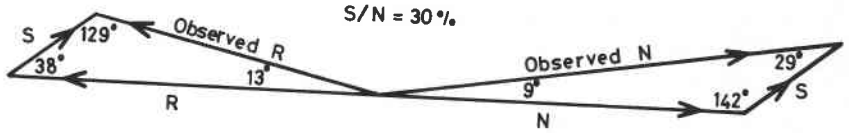


FIG. 6. Vectorial removal of secondary magnetization (S) to obtain prior magnetization direction, with normal and reversed directions antiparallel.

reverse one set and calculate the parameters of the resulting unipolar distribution. This procedure assumes that the primary and secondary magnetizations are orthogonal, which is demonstrably wrong in this instance. The results of such a calculation are given in Table 2 (entries 5, 6). Comparison of the standard method and the improved treatment employed here shows that the latter is a definite improvement in that it avoids a bias in the mean, it increases the precision of the mean, and it equalizes the precisions of the normal and reversed groups.

Application of a dip correction to these 18 site means causes no significant change in precision (Table 2, entries 17, 18), so that nothing can be said from this about the pre- or postfolding nature of this magnetization. Nevertheless, the group 1 magnetization is clearly an overprint on the group 2 magnetization and the latter does include reversals. The simplest interpretation of these observations is that the group 2 magnetization is primary.

The initial site mean directions after AF cleaning form a convincing group. The existence of two magnetizations has been revealed only by thermal demagnetization experiments and consideration of the vectors removed by cleaning.

Reversals and paleomagnetic poles

The polarity of magnetization of a site is often recoverable despite partial remagnetization. Examination of Table 1 reveals at least seven reversals in the B and C sections. Of the sites to which a polarity could be unambiguously assigned, nine are normal and nine reversed. This suggests that the two polarity states had similar probabilities, which is at variance with the conclusions of Irving and McGlynn (1976) that normal polarity is favoured. The widespread occurrence of the Coronation overprint, which has normal polarity, may be responsible for this bias.

The paleomagnetic poles derived from this study are shown in Fig. 7 together with results from rocks of similar age from other parts of the Canadian Shield. The pole associated with the group 2 magnetization (62°W, 7°S, K = 15, A_{95} = 9°) is likely to have an age lying between 1.8 Ga (age of the underlying Seton Formation) and 1.7 Ga (age of uplift). The second date is likely to apply to the group 1 magnetization. Both poles fall among other results of similar ages.

The Et-then Group gives a pole a little younger than our group 2 pole. It was sampled on Et-then Island, which is adjacent to Keith Island, but higher up in the stratigraphic column (Irving *et al.* 1972). The results from these samples fall into a normal (D = 117°, I = 33°, k = 152, N = 4) and a reversed (D = 292°, I = −16°, k = 15, N = 10) group. The groups do not have the same dispersion nor are they antiparallel. Examination shows that a very similar situation to that shown in Fig. 6 prevails, i.e., a partial remagnetization in the general direction of the Kahochella group 1 magnetiza-

TABLE 5. Site mean directions after thermal cleaning and vector subtraction of partial remagnetization, groups 2n and 2r

Site	D_p	I_p	D_{cp}	I_{cp}	D_{ca}	I_{ca}	Group
CA	166	28	168	19	166	24	2n
CB	146	22	145	14	144	35	2n
CJ	90	48	80	40	77	9	2n
CK	132	32	128	24	123	21	2n
CL	130	27	127	17	123	9	2n
CM	142	29	139	20	135	16	2n
CT	146	32	145	23	139	20	2n
CV	137	38	134	31	128	32	2n
CW	135	40	132	33	122	33	2n
BA	282	−4	290	−21	284	−14	2r
BH	286	−39	294	−47	290	−30	2r
BK	330	−14	330	−27	326	−20	2r
BM	338	9	338	−9	337	−1	2r
BN	333	−41	332	−46	327	−37	2r
CI	306	22	310	2	312	12	2r
CN	300	−2	306	−19	302	−12	2r
CO	265	−20	300	−31	294	−20	2r
CU	320	−26	322	−36	315	−39	2r

D_p, I_p = declination and inclination of site mean after thermal demagnetization, referred to present horizontal.

D_{cp}, I_{cp} = declination and inclination after vector subtraction of deduced remagnetization component (see Fig. 7).

D_{ca}, I_{ca} = declination and inclination after vector subtraction, referred to ancient horizontal.

Group: the site retains an older normally magnetized (2n) or reversely magnetized (2r) component.

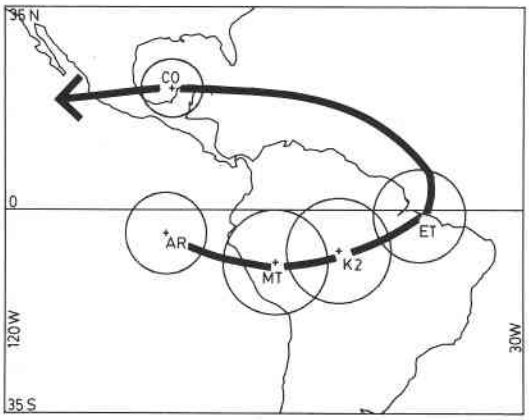


FIG. 7. Paleomagnetic poles: AR = Akaitcho River Formation (Evans *et al.* 1980); MT = Martin Formation (Evans and Bingham 1973); K2 = Kahochella group 2 (herein); ET = Et-then Group (Irving *et al.* 1972); CO = Coronation overprint (Evans *et al.* 1980). The Coronation overprint pole represents a summary of data from several sources, but is dominated by the Kahochella group 1 result reported in this paper.

tion is present. Since this overprint is strongly represented in the Kahochella samples it is not surprising to find traces of it in the nearby Et-then Group. A correction of the type shown in Fig. 6 would move the Et-then pole a little further out into the Atlantic Ocean.

As pointed out above, the group 1 overprint has been recognised and reported by a number of workers; Evans *et al.* (1980) refer to it as the Coronation overprint. It is widespread in the Athapuscow Aulacogen and corresponds to a paleomagnetic pole at 91°W, 21°N (N = 75 sites, K = 14, A_{95} = 5°, see Evans *et al.* 1980). Its probable correlatives have been recorded from other parts of the Canadian Shield, and even as far east as Greenland (see McGlynn and Irving 1978; Irving and McGlynn 1979). These secondary poles define a westerly trending portion of the Laurentian apparent polar wonder (APW) path running through the Caribbean, and for the most part, seem to correspond to uplift and cooling following the Hudsonian Orogeny, although as McGlynn and Irving (1978, p. 651) point out, orogeny in the Coronation Geosyncline probably took place somewhat later than elsewhere. Regardless of the precise sequence of events it is clear that the long-recognised widespread orogeny reflected in radiometric data has left a corresponding signature in the paleomagnetic record. Indeed, the recognition and removal of this magnetic overprint and the consequent isolation of earlier magnetizations have been major difficulties in the study of the ancient geological formations involved.

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